GENERIS*

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ABSTRACT

Here goes the abstract

1. Introduction

In recent years there has been a global expansion of renewable power generation Shahbaz, Siddiqui, Siddiqui, Jiao and Kautish (2023); Hassan, Algburi, Sameen, Al-Musawi, Al-Jiboory, Salman, Ali and Jaszczur (2024), driven by de-carbonization efforts and declining costs in renewable technologies Jäger-Waldau (2025); Joshi, Rao and Banerjee (2025); Manservigi, Castorino, Spina, Venturini and Gonzalez-Salazar (2026). This rapid deployment of renewable generation has been mostly due to additions in solar (at both utility and residential scale), hydro and wind capacity Shahbaz et al. (2023); Hassan et al. (2024). Currently, annual renewable capacity additions outpace conventional capacity additions IRENA (2025), and this upward trend will likely continue in the forecoming decades Agency (2019); Raturi (2021).

However, the rapid expansion of renewable generation is leading to grid integration difficulties due to inherent variability and uncertainty in the output Li, Shi, Cao, Wang, Kuang, Tan and Wei (2015); Liu and He (2023). Spatiotemporal mismatches between power supply and demand may arise if capacity planning and grid infrastructure development do not keep pace Ming, Kun and Jun (2013); Bird, Lew, Milligan, Carlini, Estanqueiro, Flynn, Gomez-Lazaro, Holttinen, Menemenlis, Orths et al. (2016); O'Shaughnessy, Cruce and Xu (2020); Li, Wang, Guo, Xin, Zhang, Liu and Feng (2024). Oversupply of renewable generation — occurring during periods of low demand or high wind and solar resource availability — can result in curtailment of a fraction of the generated energy Denholm, O'Connell, Brinkman and Jorgenson (2015); Bird et al. (2016); O'Shaughnessy et al. (2020). This issue is further compounded by the increasing penetration of residential-scale distributed generation, which is generally excluded from centralized dispatch scheduling owing to its negligible individual capacity Li et al. (2015); Bhattarai, Maraseni and Apan (2022). As a result, these challenges place increasing demands on power system flexibility and energy storage capacity, both of which are essential for balancing supply and demand in high-penetration renewable energy systems Denholm and Mai (2019); **?**); Nejla, Saidani and Besbes (2025).

In particular, South America has also experienced accelerated penetration of renewable generation in recent years, with wind and solar photovoltaic capacity accounting for more than half of annual capacity additions International Energy Agency (2023). Although renewable penetration has been uneven across countries, several are already experiencing increasing curtailment rates von Papp, Moraga and Henríquez (2022); Silva, Renolphi, Vieira, Lourenço, Salles and Monaro (2025). Chile has recorded curtailment rates of up to 20% of total annual renewable generation, driven mainly by a geographical mismatch between solar generation — concentrated in the northern Atacama region — and demand centers located farther south von Papp et al. (2022); Silva et al. (2025). In Brazil, by contrast, the rapid expansion of distributed generation has reshaped the load profile, also resulting in high curtailment rates da Ponte, Rego and Júnior (2025); Silva, Vieira, Rego, Simone, Lourenço and Salles (2026). Despite these challenges, the region's abundant hydroelectric capacity Farfan and Breyer (2017); International Energy Agency (2023) represents an additional advantage for renewable energy integration, offering flexible storage and dispatch capabilities that can help mitigate variability and curtailment Wan and Parsons (1993); Hingray, Favre, Sauquet and Anquetin (2014).

Bolivia's current power mix is dominated by hydro and thermal generation—accounting for approximately 70% and 20% of installed capacity, respectively — with only a small share of non-hydro renewables Autoridad de Fiscalización de Electricidad y Tecnología Nuclear (AETN) (2026) —. Efforts are underway to further decarbonize the electricity sector: national development plans Viceministerio de Electricidad y Energías Renovables (2025) and Bolivia's Nationally Determined Contributions (NDC) Ministerio de Medio Ambiente y Agua and Autoridad Plurinacional de la Madre Tierra (2021) foresee substantial growth in centralized renewable capacity. The accelerated uptake of distributed solar PV observed in the region Chueca, Weiss, Celaya, Ravillard, Ortega, Tolmasquim and Hallack (2023) suggests that distributed generation growth represents a plausible complementary pathway.

Given this growing complexity in power systems, energy modeling has become a crucial tool for exploring

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supply and demand scenarios that support long-term policy decisions Huntington, Weyant and Sweeney (1982); Pfenninger, Hawkes and Keirstead (2014). However, few studies have modeled the impacts of renewable energy penetration on power systems in South America. In Bolivia, Fernandez-Vazquez et al. Fernandez Vazquez, Vansighen, Fernandez Fuentes and Quoilin (2022) developed a long-term optimization model of the Bolivian energy sector using OSeMOSYS, based on a trend-based projection of power demand, and analyzed policy-based scenarios to induce decarbonization of the power system. However, the specific impacts of renewable capacity expansion — particularly distributed generation — on transmission requirements, thermal fuel consumption, greenhouse gas (GHG) emissions, and energy curtailment remain unexplored. This study aims to analyze the impacts of exogenous renewable capacity additions — both centralized and distributed — on power transmission, fuel consumption in thermal power facilities, GHG emissions, and energy curtailment. An intraday hydroelectric dispatch strategy is also proposed to mitigate renewable energy curtailment. For this purpose, we constructed a regionalized model of the Bolivian energy sector using the LEAP and NEMO modeling tools.

The structure of the paper proceeds as follows: Section 2...

2. Methodology

2.1. Case study

Electricity demand currently represents approximately 11% of Bolivia's overall energy demand and has grown at an annual rate of approximately 5.7% in recent years **??**. This demand is predominantly met by natural gas (NG) and hydroelectric generation. By the end of 2024, the installed capacity of the National Interconnected System (NIS) totaled 3580 MW, of which 2573 MW corresponded to thermoelectric sources and 758 MW to hydroelectric sources, with the remaining 249 MW distributed across wind, solar, and biomass. Approximately 98% of thermoelectric capacity is NG-based, with the remainder corresponding to diesel engines; the NG-based fleet, in turn, is composed of 66% combined-cycle, 30% turbo-gas, and 4% turbo-vapor units.

NG-based generation accounts for most of Bolivia's capacity expansion over the last three decades **?**. More recently, however, a steep decline in domestic NG production **?** has motivated the development of national plans and policies aimed at decarbonizing the power sector **?**Fernandez Vazquez et al. (2022). The capacity expansion plan assessed in this study, prepared by the Ministry of Energy, includes two ambitious scenarios of 100% renewable capacity additions up to 2037.

Bolivia's NIS has no current linkages with the national grids of bordering countries and can therefore be considered an isolated system; it currently covers eight out of nine departments. Given this configuration, Bolivia's energy system was represented in this study as a multi-nodal system, consisting of four nodes or regions obtained by grouping the administrative divisions as follows:

Table 1
Power lines

Line	Nodes	Capacity [MW]
L1	Centro-Norte	390
-L2	Centro-Sur	260
-L3	Centro-Oriente	165
L4	Oriente-Norte	35
-L5	Centro-Oriente	500

- Norte (La Paz + Beni)
- Centro (Cochabamba + Oruro)
- Oriente (Santa Cruz)
- Sur (Tarija, Potosí, and Chuquisaca)

This regionalization was based on prior energy modeling studies conducted in Bolivia that proposed this specific departmental aggregation (see **?**). Accordingly, this structure was replicated in the present study to facilitate comparison with independently conducted modeling exercises.

Figure 1 shows the location of the nodes and the representation of the NIS, for which two power lines connecting Centro and Oriente, one line connecting Centro and Sur, one line connecting Centro and Norte, and one connecting Oriente and Norte were considered. The main features of the power lines are detailed in Table 1.

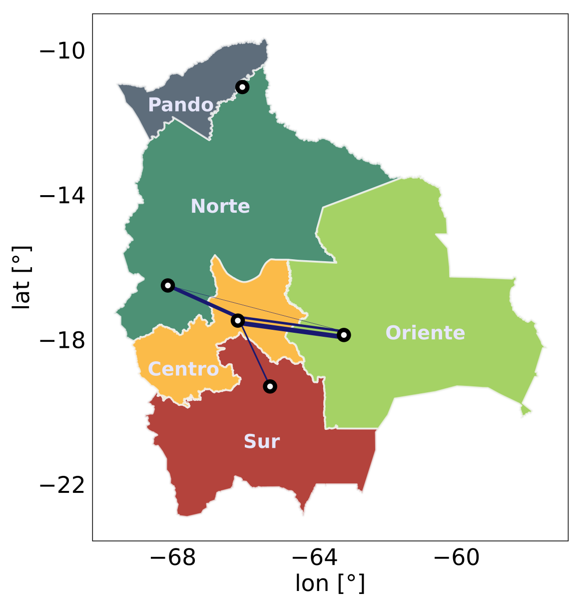

Figure 1: Regional nodes included in the model and power lines

Figure 2:

Table 2

3.

CRediT authorship contribution statement

Emilio Bianchi: Formal analysis, Investigation, Methodology, Visualization, Writing 1–original draft. **Ignacio Sagardoy:** Conceptualization, Formal analysis, Investigation, Methodology, Validation. **Francisco Lallana:** Conceptualization, Funding acquisition, Investigation, Methodology, Supervision, Validation. **Gustavo Nadal:** Conceptualization, Funding acquisition, Investigation, Methodology, Supervision. **Gonzalo Bravo:** Funding acquisition, Project administration. **Gonzalo Bravo:** Conceptualization of this study, Methodology, Software.

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